

Assessment of Sensorimotor Complications After Plaque Brachytherapy for Patients With Uveal Melanoma



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- **OBJECTIVE:** To evaluate the incidence and characterization of diplopia and strabismus following treatment with Iodine-125 plaque brachytherapy for uveal melanoma.
- **DESIGN:** Retrospective single-center clinical case-control study.
- **PARTICIPANTS:** Patients with uveal melanoma treated with Iodine-125 plaque brachytherapy between May 1, 2014 and December 1, 2022.
- **METHODS:** A chart review was conducted for patients with uveal melanoma treated with plaque brachytherapy. Data on demographics, tumor characteristics, surgical technique, radiation dosing, and sensorimotor complications were collected.
- **MAIN OUTCOME MEASURES:** Data was analyzed to assess the incidence, characteristics, treatment, and outcomes of diplopia and/or strabismus following plaque brachytherapy. Statistical analysis was performed to assess for associations between sensorimotor complications and independent variables.
- **RESULTS:** A total of 375 patients (mean age 64 ± 14 years; 50.4% female) were treated with Iodine-125 plaque brachytherapy. Mean follow-up time was 3.68 ± 2.3 years. A total of 321 muscles were manipulated for plaque placement, most commonly the lateral rectus (42.1%; 135/321) followed by the superior rectus (22.4%; 72/321) and inferior rectus (17.4%; 56/321). This was done most often through disinsertion of the muscle (96.3%; 309/321). Postoperative diplopia occurred in 38.9% of patients (146/375). Diplopia typically lasted less than 6 months (58.2%; 85/146) and spontaneous resolution was seen in most patients (71.2%; 104/146). In total, 7.2% (27/375) of the patients had documentation for type of strabismus, with a horizontal component noted most often (74.1%; 20/27). The incidence of sensory-deprivation related strabismus was

3.7% (14/375). In a multivariate regression model, there were no factors associated with increased odds of developing diplopia. Those with worse baseline visual acuity and greater tumor distance from the optic disc had lower odds of diplopia ($P = .005$ and $P = .026$, respectively). Superior rectus muscle manipulation ($P = .010$), younger age ($P = .045$), and moderate ($P = .022$) or severe ($P = .045$) vision loss were associated with increased odds of developing strabismus.

- **CONCLUSIONS:** Iodine-125 plaque brachytherapy for uveal melanoma carries a moderate risk of diplopia, which typically resolves on its own. Sensory deprivation-related strabismus may also occur. These findings highlight that although sensorimotor complications may occur, they often resolve within 6 months postoperatively with no intervention. (Am J Ophthalmol 2026;281: 624–630. © 2025 Elsevier Inc. All rights are reserved, including those for text and data mining, AI training, and similar technologies.)

UVEAL MELANOMA IS THE MOST COMMON PRIMARY intraocular malignancy in adults, with an incidence of 1500 cases per year in the United States.^{1,2} Treatment aims to achieve tumor regression while trying to preserve the eye and vision, most commonly through radiotherapy.^{3–7} Plaque brachytherapy involves suturing a radioactive plaque to the sclera overlying the tumor and may require manipulation of ocular muscles to ensure proper positioning. This can include partial or complete disinsertion of a muscle often with re-insertion of the muscle at the time of plaque removal. Occasionally, the plaque can be placed under the muscle.^{8–10} While effective, brachytherapy carries unique risks beyond typical radiation-induced complications, notably strabismus and diplopia.^{11–15} This may be secondary to radiation damage, mechanical injury, or loss of sensory fixation.¹⁶

Reported incidence rates of strabismus after plaque brachytherapy range widely (2.5% to 60%) with limited research on late-onset or persistent strabismus.^{17,18} The literature shows inconsistent or conflicting findings on predisposing risk factors for the development of sensorimotor complications.^{19,20} Furthermore, many of these studies have been limited by small sample sizes. This study

Accepted for publication October 11, 2025.

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